# COMMON QUARTETS OF SEVERAL PHYLOGENETIC TREES: $h(3) = 14$ AND EXPONENTIAL BOUNDS FOR $h(k)$

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Abstract. Let $h(k)$ be the least $n$ such that any $k$ unrooted binary phylogenetic trees on the same $n$ leaves display a common quartet. Snir and Yuster asked for its value; it was known that $h(2) = 6$ and $11 \le h(3) \le 14$, and the best published upper bound for general $k$ was a four-times iterated exponential. We show $h(3) = 14$ and $h(k) \le 2 \cdot 3^{k-1} + 1$ for every $k \ge 2$. The upper bound applies to trees the iteration of Spencer's two-bipartition lemma that Asplund, Do, Hamm, Székely, Taylor and Wang used for $k$-planar crossing numbers. Snir and Yuster stated that caterpillars give $h(k) \ge 2^k + 2$ and omitted the construction; we give an explicit family of $k$ caterpillars on $2^k + 1$ leaves, built by reflecting leaf orders according to binary digits, with a proof. The value $h(3) = 14$ needs three trees on 13 leaves with no common quartet; we give one of the form $T$, $3T$, $9T$ on $\mathbb{Z}_{13}$, and confirm the matching upper bound by an exhaustive search with checked proofs. Up to the symmetries of this form there are exactly six such examples. We also find four trees on 21 leaves with no common quartet, so $h(4) \ge 22$. The general bounds and the 13-leaf example are checked in Lean.

## 1. Introduction

Given $k$ evolutionary trees on the same species, how many species force them all to agree on the shape of some four-species subtree? This note answers the question for three trees and narrows it for every $k$.

A phylogenetic tree on a leaf set $X$ is an unrooted tree whose internal vertices have degree three and whose leaves are labelled bijectively by $X$. Four leaves $a, b, c, d$ induce one of three binary trees, written $ab|cd$, $ac|bd$ or $ad|bc$; this is the *quartet* the tree displays on $\{a, b, c, d\}$, and $ab|cd$ means that some edge has $a, b$ on one side and $c, d$ on the other. A tree is determined by its quartets [4, 2]. Comparing trees through their quartets is standard in phylogenetics, and a basic extremal question is how different several trees on the same species can be. Two random trees agree, in expectation, on a third of their quartets, yet two trees can avoid agreeing altogether when there are few leaves. Figure 1 shows two trees on five leaves with no quartet in common. On six or more leaves this is impossible: any two trees share a quartet [2].

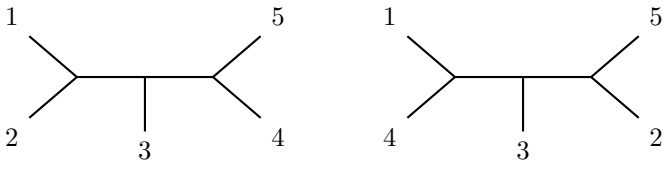



Figure 1. Two trees on five leaves with no common quartet. Each quartet is the four leaves left after deleting one. Deleting 5 gives 12|34 in the left tree and 14|23 in the right; deleting 1 gives 23|45 and 25|34; the other three quartets differ in the same way. The right tree reverses the middle three leaves of the left one, the move that drives the construction in Section 4.

Snir and Yuster [11, §5] asked for the least $n$ such that any $k$ phylogenetic trees on $[n]$ display a common quartet, and denoted it $h(k)$. The result of Bandelt and Dress gives $h(2) = 6$. For caterpillars (trees in which every internal vertex lies on one path) Snir and Yuster stated that

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the analogous number is $h^*(k) = 2^k + 2$, proving the upper bound and giving the construction for the lower bound only for $k = 3$. Three trees on ten leaves with no common quartet [11, Figure 4] and their separation theorem [11, Theorem 1.3] give $11 \le h(3) \le 14$. Currie, Czabarka, Fischer, Simmons, Székely and Wicke [6] proved $8^{1/4}3^{k/4} \le h(k) \le \mathrm{tw}_3\big(6(k-1)2^k + O(\log k)\big)$, a four-times iterated exponential, the upper bound through hypergraph Ramsey numbers and agreement subtrees.

**Theorem 1.1.** $h(3) = 14$.

**Theorem 1.2.** *For every* $k \ge 2$, $\quad 2^k + 2 \le h(k) \le 2 \cdot 3^{k-1} + 1$.

**Theorem 1.3.** $22 \le h(4) \le 55$, *and* $h(4) \le 41$ *given the error-free form of* [11, Theorem 1.3].

The lower bound in Theorem 1.2 is $h(k) \ge h^*(k)$, since caterpillars are trees, and the value $h^*(k) = 2^k + 2$ is Snir and Yuster's; Section 4 supplies the construction they omitted, an explicit family of $k$ caterpillars on $2^k + 1$ leaves with no common quartet, with a short proof. The upper bound (Section 3) iterates a lemma of Spencer, as Asplund et al. [1] did for $k$-planar crossing numbers: each further tree costs a factor of three in the size of two sets that every tree separates, and two such sets of size two already give a common quartet. Starting the iteration from Snir and Yuster's three-tree theorem gives $h(k) \le \lfloor 3^k/2 \rfloor + 1$ for $k \ge 3$, which is 14 for $k = 3$ and 41 for $k = 4$. The proof of that theorem without its error term is given in [11] in outline, so we state the bounds that rest on it separately.

For three trees the lower bound $2^3 + 2 = 10$ is far from the truth. Section 5 gives three trees on 13 leaves with no common quartet, in a symmetric form: a single tree $T$ on $\mathbb{Z}_{13}$ together with its images under $x \mapsto 3x$ and $x \mapsto 9x$. The upper bound $h(3) \le 14$ is Snir and Yuster's, whose proof of the error-free form of their separation theorem is a sketch; we confirm it independently by an exhaustive SAT search over the 135 shapes of the first tree, each unsatisfiable instance carrying a checked proof.

The gap between $2^k$ and $3^k$ is the main question left open (Section 6). The data $h(2) = 6$, $h(3) = 14$, $h(4) \ge 22$ beat the caterpillar values $6, 10, 18$ from $k = 3$ on.

## 2. Separated sets

For a tree $T$ on $X$ and disjoint $A, B \subseteq X$, say $T$ *separates* $A$ and $B$ if some edge of $T$ has $A$ on one side and $B$ on the other. Then $T$ displays $ab|cd$ exactly when it separates $\{a, b\}$ and $\{c, d\}$, and if $T$ separates $A$ and $B$ it separates any subsets of them. For $U \subseteq X$ the restriction $T|U$ is the phylogenetic tree obtained from the subtree spanning $U$ by suppressing vertices of degree two. An edge of $T|U$ comes from a path of $T$, and deleting any edge of that path splits $U$ the same way; so a split $C|U \setminus C$ of $T|U$ means that $T$ separates $C$ and $U \setminus C$.

**Lemma 2.1** (folklore; see [10, Lemma 3.1])**.** *Every phylogenetic tree on* $N \ge 2$ *leaves has an edge whose removal leaves at least* $N/3$ *leaves on each side.*

*Proof.* Choose an edge $e$ whose larger side $S$ has as few leaves as possible, and suppose $|S| > 2N/3$. Then $|S| \ge 2$, so the end of $e$ in $S$ is an internal vertex, and its two other edges split $S$ into $S_1 \sqcup S_2$ with, say, $|S_1| \ge |S|/2 > N/3$. The edge leading to $S_1$ has sides of sizes $|S_1| < |S|$ and $N - |S_1| < 2N/3 < |S|$, contradicting the choice of $e$. So $|S| \le 2N/3$ and the other side has at least $N/3$ leaves. □

**Lemma 2.2** (Spencer [12]; see [1, Lemma 6] and [6, Theorem 3])**.** *Let* $U = X_1 \sqcup X_2 = C_1 \sqcup C_2$ *be two bipartitions with every part of size at least* $|U|/3$*. Then either* $X_1 \cap C_1$ *and* $X_2 \cap C_2$*, or* $X_1 \cap C_2$ *and* $X_2 \cap C_1$*, both have size at least* $|U|/6$*.*

*Proof.* Call a cell small if it has fewer than $|U|/6$ elements. If the lemma fails, each of the two pairs of opposite cells contains a small cell. Any cell of the first pair shares a row or a column with any cell of the second, so two small cells lie in one part, which then has fewer than $|U|/3$ elements. □

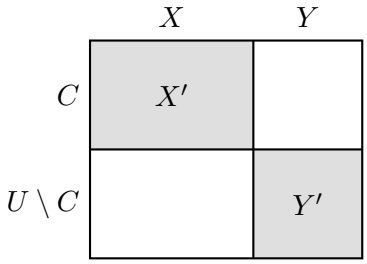



FIGURE 2. One step of the iteration. The earlier trees separate $X$ and $Y$; the new tree, restricted to $U = X \cup Y$, splits $U$ into $C$ and $U \setminus C$. Lemma 2.2 finds opposite cells $X'$, $Y'$ of at least a sixth of $U$, and every tree separates them.

## 3. The upper bound

The following proposition is the iteration in the proof of Theorem 2 of Asplund et al. [1], where the bipartitions are 1/3–2/3 bisections of graphs; here they come from tree edges through Lemma 2.1. We include the short proof to keep the note self-contained.

**Proposition 3.1.** *Let $k \geq 2$ and let $T_1, \dots, T_k$ be phylogenetic trees on a set of $n$ leaves. There are disjoint sets $X, Y$ separated by every $T_i$ with $2 \cdot 3^{k-1}|X| \geq n$ and $2 \cdot 3^{k-1}|Y| \geq n$.*

*Proof.* Induction on $k$. For $k = 2$, Lemma 2.1 gives a split $X_1|X_2$ of the leaf set displayed by $T_1$ and a split $C_1|C_2$ displayed by $T_2$, all four parts of size at least $n/3$; Lemma 2.2 gives opposite cells of size at least $n/6$, separated by both trees.

Suppose $X, Y$ work for $T_1, \dots, T_k$. Shrink them to subsets $X_0, Y_0$ of a common size $m = \min(|X|, |Y|)$ and let $U = X_0 \cup Y_0$, so $|U| = 2m$. By Lemma 2.1 applied to $T_{k+1}|U$, the tree $T_{k+1}$ separates some $C$ and $U \setminus C$ with $|C|, |U \setminus C| \geq 2m/3 = |U|/3$. Both parts of $X_0 \sqcup Y_0$ have size $m \geq |U|/3$, so Lemma 2.2 gives $X' \subseteq X_0$ and $Y' \subseteq Y_0$, one inside $C$ and the other inside $U \setminus C$, each of size at least $|U|/6 = m/3$ (Figure 2). Then $T_{k+1}$ separates $X'$ and $Y'$, the earlier trees do because $X' \subseteq X$ and $Y' \subseteq Y$, and $2 \cdot 3^k|X'| \geq 2 \cdot 3^{k-1}m \geq n$. □

*Proof of the upper bounds in Theorems 1.2 and 1.3.* If $n \geq 2 \cdot 3^{k-1} + 1$ then $|X|, |Y| > 1$ in Proposition 3.1. Choose $a, b \in X$ and $c, d \in Y$: every tree separates $\{a, b\}$ and $\{c, d\}$, so every tree displays $ab|cd$. For $k = 4$ this gives 55. For $k \geq 3$, starting the induction at $k = 3$ from the error-free form of [11, Theorem 1.3], which gives separated sets with $27|X|, 27|Y| \geq 2n$, yields $\frac{1}{2}3^k|X| \geq n$ and hence $h(k) \leq \lfloor 3^k/2 \rfloor + 1$; for $k = 4$ this is 41. □

## 4. Caterpillars on $2^k + 1$ leaves with no common quartet

A caterpillar is determined, up to reversal, by the order in which its leaves meet the spine, and the caterpillar with order $x_1, \dots, x_n$ displays $x_p x_q | x_r x_s$ whenever $p < q < r < s$. We describe a caterpillar by a *position map*: a bijection from the leaves to $\mathbb{Z}$ whose order is the leaf order.

**Definition 4.1.** Let $k \geq 1$ and $h = 2^{k-1}$. The leaves are the integers $-h, \dots, h$. The caterpillar $\pi_0$ has the identity position map. For $0 \leq b \leq k-2$, the caterpillar $\pi_{b+1}$ puts leaf $x$ at position $-x$ if $x \neq 0$ and the binary digit of $|x| - 1$ in place $b$ is 0, and at position $x$ otherwise. Let $F_k = \{\pi_0, \dots, \pi_{k-1}\}$.

Each $\pi_j$ swaps some pairs $\{-d, d\}$ and fixes the rest, so it is a bijection of $\{-h, \dots, h\}$ fixing 0. For $k = 3$ the three leaf orders are

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| $\pi_0$ : | $-4$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ |
| $\pi_1$ : | $-4$ | $3$ | $-2$ | $1$ | $0$ | $-1$ | $2$ | $-3$ | $4$ |
| $\pi_2$ : | $-4$ | $-3$ | $2$ | $1$ | $0$ | $-1$ | $-2$ | $3$ | $4$ |

and any four of the nine leaves are split differently by at least two of them. Relabelling $x \mapsto x+5$ turns these into the three orders 123456789, 127654389, 183654729 of Snir and Yuster's example [11, §5] (the first is printed there as 12345679), so $F_k$ extends their construction to every $k$.

**Theorem 4.2.** *No quartet is displayed by all the caterpillars of $F_k$.*

*Proof.* Induction on $k$; for $k = 1$ there are three leaves. Assume the result for $F_k$, with $h = 2^{k-1}$, and consider $F_{k+1}$ on $[-2h, 2h]$. Write $I = [-h, h]$ for the inner leaves and $L = [-2h, -h-1]$, $R = [h+1, 2h]$. Let $\pi_{\text{top}} = \pi_k$ be the caterpillar of the highest digit. Four facts follow from the definition (Figure 3).

(1) On $I$, the caterpillars $\pi_0, \dots, \pi_{k-1}$ of $F_{k+1}$ have the same position maps as those of $F_k$.
(2) For an outer leaf $x$ let $\psi(x) = x - h$ if $x > 0$ and $x + h$ if $x < 0$, so $\psi$ maps $L \cup R$ onto $I \setminus \{0\}$. The digits of $|x| - 1$ and $|\psi(x)| - 1$ agree in places below $k-1$, so for $j < k$ the position of $x$ in $\pi_j$ is $g$(position of $\psi(x)$ in $\pi_j$), where $g(y) = y + h$ for $y > 0$ and $y - h$ for $y < 0$; and $g$ is increasing.
(3) $\pi_{\text{top}}$ reverses $I$ and fixes every outer leaf.
(4) In every caterpillar, inner leaves sit at positions in $[-h, h]$ and outer leaves at positions outside it.

Suppose all of $F_{k+1}$ display $ab|cd$, and let $Q = \{a, b, c, d\}$.

*At most one leaf of $Q$ is inner.* Map each outer leaf by $\psi$ and the inner leaf, if any, to 0. By (2) and (4) the relative order of the four positions in $\pi_j$ ($j < k$) is the relative order of the images in the $\pi_j$ of $F_k$, so $F_k$ displays a common quartet, contrary to induction.

*All four leaves are inner.* By (1) and induction.

*Three inner leaves and one outer leaf $y$.* In $\pi_0$ and $\pi_{\text{top}}$ the leaf $y$ is at the same end, and the inner three are in opposite orders; so $y$ is paired with different inner leaves.

*Two inner leaves $u < v$ and two outer leaves $y, z$.* If $y$ and $z$ lie on opposite sides in $\pi_0$, then $\pi_0$ pairs each with the nearer of $u, v$, and $\pi_{\text{top}}$, which swaps $u$ and $v$, pairs them the other way. If they lie on the same side then $|y| \neq |z|$, so $|y| - h - 1$ and $|z| - h - 1$, two different numbers below $h = 2^{k-1}$, differ in some place $b < k-1$. Then $\pi_{b+1}$ moves exactly one of $y, z$ to the other side, and it pairs each of them with an inner leaf, while $\pi_0$ displays $yz|uv$. □

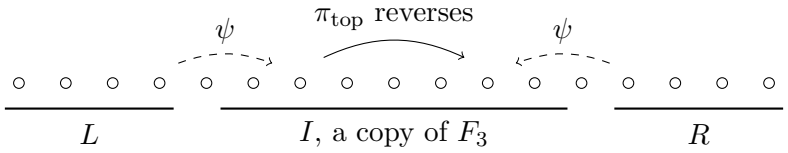



FIGURE 3. The doubling step from $F_3$ to $F_4$. The inner nine leaves carry $F_3$; the outer eight form a copy of $F_3$ without its centre, folded in by $\psi$; the new caterpillar reverses the inner block and fixes the rest.

**Corollary 4.3.** *$h(k) \geq h^*(k) = 2^k + 2$ for every $k \geq 1$.*

The upper bound $h^*(k) \leq 2^k + 2$ is Snir and Yuster's halving argument [11, §5]. The choice of digit place is not essential: using the digits of $|x|$ instead of $|x| - 1$ gives another family with the same property for $k \leq 4$, which we checked by computer.

## 5. Three trees

**Proposition 5.1.** *Let $T$ be the phylogenetic tree on $\mathbb{Z}_{13}$ with Newick form*

$$\big(0,\ 1,\ (((8,11),((5,(4,6)),((3,9),(12,(7,10))))),\ 2)\big),$$

*and let $3T$ and $9T$ be its images under $x \mapsto 3x$ and $x \mapsto 9x$. No quartet is displayed by all of $T$, $3T$ and $9T$.*

The proof is a finite check of the $\binom{13}{4} = 715$ quartets, done in Lean by `decide` and by an independent script (Section 7). The example was found by a SAT search in which the second and third trees were forced to be the images of the first under a permutation of order three; without the symmetry constraint the search also finds examples at once. Section 5.1 describes every example of this symmetric form; we do not know a conceptual reason why they work.

*Proof of Theorem 1.1.* Proposition 5.1 gives $h(3) \geq 14$, and Theorem 1.3 of [11] gives $h(3) \leq 14$, since $\lceil 2 \cdot 14/27 \rceil = 2$. Because the error-free form of that theorem is proved there in outline, we confirmed $h(3) \leq 14$ directly. A choice of one split for every 4-subset is the quartet set of a phylogenetic tree if and only if its restriction to every 5-subset is [4] (see [7] for this formulation). This gives a propositional formula whose models are triples of trees on 14 leaves with no common quartet. A permutation of the leaves applied to all three trees preserves the property of having no common quartet, so we may take the first tree to be a fixed representative of its shape; there are 135 shapes of phylogenetic trees with 14 leaves. All 135 formulas are unsatisfiable. Each was solved by CaDiCaL 2.1.3 [3], which wrote an LRAT proof of unsatisfiability, and each proof was accepted by the independent checker `lrat-check` [5]. The proofs total 32.6 GB and were deleted after checking; the archive contains the scripts that regenerate and check them (about 5,500 s of solver time) and the logs of the run. □

*Proof of the lower bound in Theorem 1.3.* Let $\sigma$ be the permutation of $\{0, \dots, 20\}$ that fixes 0 and is the product of the 4-cycles

$$(1\,2\,3\,4)\,(5\,6\,7\,8)\,(9\,10\,11\,12)\,(13\,14\,15\,16)\,(17\,18\,19\,20),$$

and let $T$ be the tree

$$(0,\ 16,\ ((17,10),(1,((((((6,14),11),5),((3,15),8)),$$
$$((4,13),(18,((20,7),19)))),(9,(2,12)))))).$$

Then $T$, $\sigma T$, $\sigma^2 T$, $\sigma^3 T$ display no common quartet, checked over all $\binom{21}{4} = 5985$ quartets. The search for 22 leaves did not finish. □

5.1. **The examples on $\mathbb{Z}_{13}$.** The nonzero elements of $\mathbb{Z}_{13}$ form a group of order 12, and the orbits of $x \mapsto 3x$ on them are the four cosets

$$\{1,3,9\}, \qquad \{2,6,5\}, \qquad \{4,12,10\}, \qquad \{7,8,11\}$$

of the subgroup $\{1,3,9\}$. Write $x = 3^i c_j$ with $i \in \mathbb{Z}_3$ and $(c_0, c_1, c_2, c_3) = (1,2,4,7)$. The leaves other than 0 become a $3 \times 4$ grid whose columns are these cosets, and $x \mapsto 3x$ becomes the shift $i \mapsto i+1$ in every column, so the three trees of Proposition 5.1 are the translates of one tree along the columns. The 21-leaf example has the same shape: a fixed leaf and a $4 \times 5$ grid, with $\sigma$ shifting every column.

One family of quartets is disposed of by the symmetry alone. If $Q$ consists of 0 and a whole column, the shift maps $Q$ to itself and permutes the three column leaves cyclically, so it moves the leaf paired with 0; hence $T$, $3T$ and $9T$ display three different splits of $Q$. The other quartets need the particular tree.

The maps fixing 0 and sending $(i,j)$ to $(\pm i + s_j, \rho(j))$, with $s \in \mathbb{Z}_3^4$ and $\rho$ a permutation of the columns, commute with the shift up to its inverse, so they carry a tree with the property of Proposition 5.1 to another. They form a group of order $3^4 \cdot 24 \cdot 2 = 3888$.

**Proposition 5.2.** *Exactly* 19,440 *phylogenetic trees $T$ on $\mathbb{Z}_{13}$ are such that $T$, $3T$ and $9T$ display no common quartet. Under the group above they form six orbits: two of* 1944 *trees and four of* 3888*.*

This is a computation. A single tree on $\mathbb{Z}_{13}$ is encoded through its rooted triples with 0 as root, the condition on $T$, $3T$, $9T$ is written in the variables of $T$ alone, and every model is excluded after it is found until none remain; the orbits are then computed directly. The trees in the two smaller orbits have a symmetry of order two. One of them is

$$T' = \big((0,(2,9)),\ ((7,11),(1,(3,4))),\ ((10,12),(6,(5,8)))\big),$$

which has a central vertex with branches of 3, 5 and 5 leaves. The involution

$$\iota = (1\,6)(2\,9)(3\,5)(4\,8)(7\,10)(11\,12)$$

commutes with $x \mapsto 3x$, fixes $T'$, and exchanges its two branches of five leaves. We have not found a pattern in these examples that extends to more trees.

## 6. Open questions

**Question 6.1.** Does $\lim_{k\to\infty} h(k)^{1/k}$ exist, and is it larger than 2?

Theorem 1.2 puts any limit in $[2, 3]$. The caterpillar family is optimal among caterpillars, and the values for $k = 3, 4$ show that general trees do better. A product of two families does not help directly: if one group of trees keeps the rows of a grid of leaves together and another group the columns, the four leaves $(x, u), (x, u'), (y, v), (z, v)$ are split the same way by all of them. The caterpillar doubling works because the inner block lies between two outer zones on the spine; an analogue for general trees would insert one family along a path of another.

For $k = 3$ the bound $\lfloor 3^k/2 \rfloor + 1$ is attained. Moreover $13 = (3^3 - 1)/2$ is the number of points of the projective plane over the field with three elements, and on a Singer cycle of that plane the Frobenius map acts as $x \mapsto 3x$ on $\mathbb{Z}_{13}$. If this pattern continued, the limit would be 3. Its most direct analogue for four trees fails for a simple reason.

**Proposition 6.2.** *Let $\sigma$ be a permutation of order $k$ of the leaf set, and suppose that for some $d < k$ dividing $k$ the permutation $\sigma^d$ fixes at least $h(d)$ leaves. Then for every tree $T$ the trees $T, \sigma T, \ldots, \sigma^{k-1}T$ display a common quartet. In particular no tree $T$ on $\mathbb{Z}_{40}$ makes $T$, $3T$, $9T$, $27T$ free of common quartets.*

*Proof.* Let $F$ be the set of leaves fixed by $\sigma^d$; it is mapped to itself by $\sigma$. On $F$ the tree $\sigma^{i+d}T$ restricts to the same tree as $\sigma^i T$, so the $k$ restrictions are the $d$ trees $T|F, \ldots, (\sigma^{d-1}T)|F$ repeated. Since $|F| \geq h(d)$ these $d$ trees display a common quartet, and so do all $k$. On $\mathbb{Z}_{40}$, $x \mapsto 9x$ fixes the eight multiples of 5, and $8 \geq h(2) = 6$. □

Searches for four trees forming the orbit of one tree under a permutation of order four found examples on 17 and 21 leaves at once, and gave no answer within 10 to 20 minutes on 22, 24 and 25 leaves (with the permutation a product of 4-cycles and fixed points) or on $\mathbb{Z}_{29}$ and $\mathbb{Z}_{37}$ under multipliers of order four. Five trees on $\mathbb{Z}_{41}$ under $x \mapsto 10x$, which would beat the 33 leaves of $F_5$, were also undecided after 20 minutes. These unfinished searches decide nothing, but 21 remains the largest example known for four trees.

**Question 6.3.** What is $h(4)$? It lies between 22 and 55, and below 41 given the error-free form of [11, Theorem 1.3]. Is it 41, as the pattern of $h(3)$ would suggest?

## 7. Verification

Lean 4 with mathlib [8] checks: Proposition 3.1 and the bound $h(k) \leq 2 \cdot 3^{k-1} + 1$, with trees abstracted to split systems satisfying Lemma 2.1 and closure under subsets, so that Lemma 2.2 and the iteration are proved; Theorem 4.2 for every $k$, following the proof above; and Proposition 5.1 by `decide`, with no axioms. Each Lean file is accompanied by a mutation suite: deliberately false variants of the statements (a threshold one lower, an iteration losing only a factor of two, one caterpillar fewer, a 13-leaf tree with two labels swapped), each of which Lean rejects. Python scripts recheck the examples with code that shares nothing with the search, validate the SAT encoding (it counts 15, 105 and 945 trees on 5, 6 and 7 leaves, and gives $h(2) = 6$), and check $F_k$ directly for $k \leq 7$, that is up to 129 leaves. The one-tree encoding behind Proposition 5.2 and the searches of Section 6 is validated the same way: it also counts 15, 105 and 945 trees, finds examples on 13 and 21 leaves, and refutes the symmetric instance on 14 leaves with an LRAT proof accepted by `lrat-check`. The count 19,440 in Proposition 5.2 was repeated with the five-point encoding of Section 5, which shares no code with the one-tree encoding. The search programs, the verification scripts and the Lean formalisation were written with the AI assistant named in the declaration below, and were checked as described in this section. Code, logs and Lean sources are at [9].

## Declaration of generative AI and AI-assisted technologies in the research and writing process

During the preparation of this work the author used Claude (Anthropic; model Claude Opus 5.5) to search for examples by SAT, to find the caterpillar family and its proof and

the application of the iteration of Asplund et al. to trees, to classify the examples on $\mathbb{Z}_{13}$, to write the verification programs and the Lean formalisation, and to draft the text. The author directed the work and takes full responsibility for the content of this preprint, which has not yet been independently reviewed.

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